

jamRF-v2: High-Power Chaos-Modulated Jammer
Baseband / Frontend Architecture (HackRF One style)

GPS L1 (1.575 GHz) + Wi-Fi/ISM (2.4 GHz)

System Architecture:

- Baseband: ESP32 controller, chaos modulation DAC outputs
- Frontend GPS: VCO -> Atten -> BPF -> GaN PA -> Isolator -> SMA
- Frontend WiFi: VCO -> Atten -> BPF -> WiFi PA -> Isolator -> SMA
- Power: Battery, buck, LDO, charge pump, MOSFET switches
- Control: Thermal failsafe, bias sequencing, RF shielding

Baseband

4-Layer FR4, 0.062", 1oz Cu, 50-ohm controlled impedance

File: baseband.kicad_sch

Frontend - GPS L1

File: frontend_gps.kicad_sch

Frontend - WiFi 2.4 GHz

File: frontend_wifi.kicad_sch

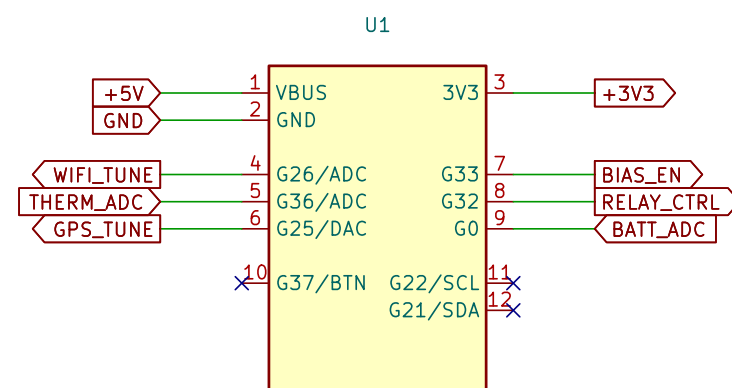
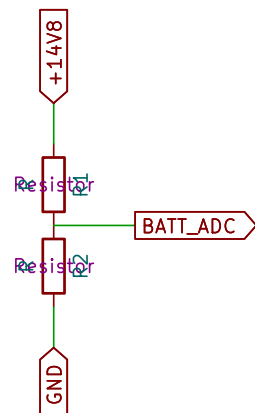
Power Distribution

File: power.kicad_sch

Control and Safety

File: control.kicad_sch

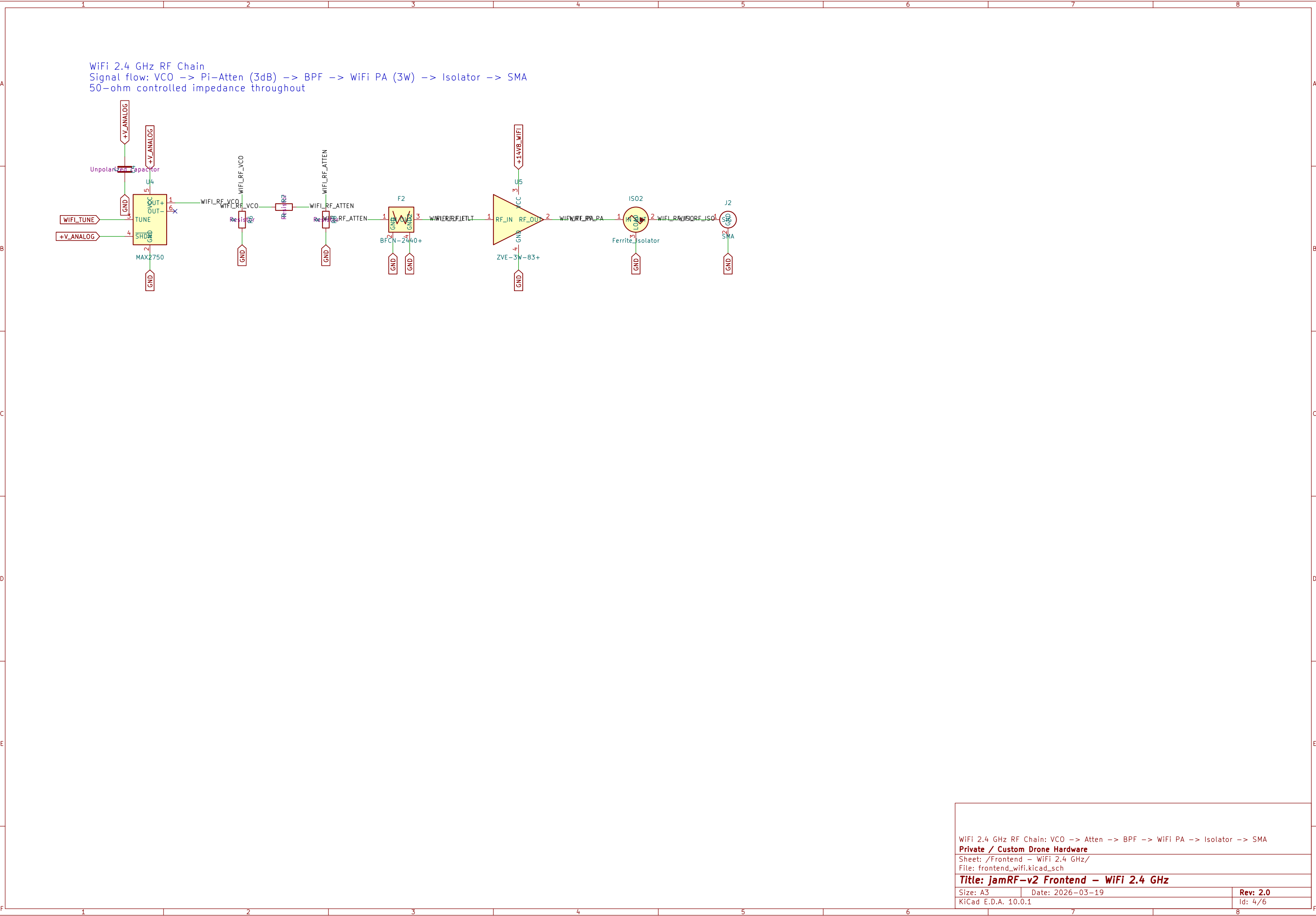
Thermal Failsafe + GaN Bias Sequencing		
14.8V 4S LiPo Power Source		
GPS L1 (1.575 GHz) + Wi-Fi/ISM (2.4 GHz)		
Top-Level Hierarchical Sheet - Baseband/Frontend Architecture		
Private / Custom Drone Hardware		
Sheet: /		
File: jamrf-v2.kicad_sch		
Title: jamRF-v2 Chaos-Modulated Jammer		
Size: A3	Date: 2026-03-19	Rev: 2.0
KiCad E.D.A. 10.0.1		Id: 1/6



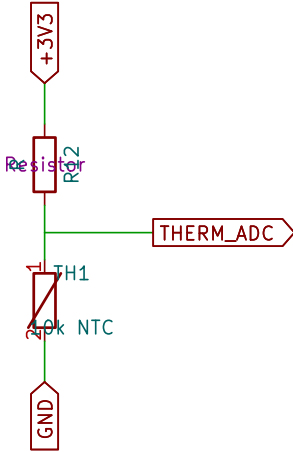
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Chaos Modulation DAC Outputs:
G25/DAC -> GPS_TUNE (VC01)
G26/DAC -> WIFI_TUNE (VC02)

Analog Inputs:
G36/ADC <- THERM_ADC (temperature sense, ADC1_CH0)
G0/ADC  <- BATT_ADC (battery voltage, via divider)

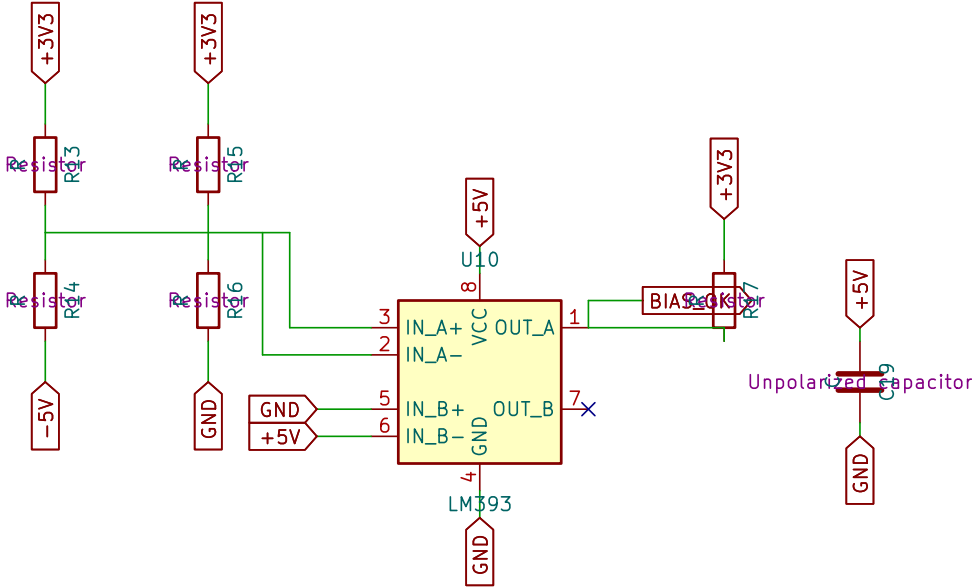
Digital Control:
G33 -> BIAS_EN (charge pump enable)
G32 -> RELAY_CTRL (WiFi PA switch)
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THERMAL MONITORING
+3V3 -> R_TH (10k fixed) -> midpoint -> TH1 (10k NTC) -> GND
Midpoint to THERM_ADC -> G36 (ADC1_CH0, always available)
 $R_{ntc} = R_{TH} * adc / (ADC_MAX - adc)$
At 25C: ~1.65V At 85C: ~2.7V (voltage rises with temp)



BIAS SEQUENCING (sense on IN_A-, ref on IN_A+)
Sense divider: +3V3 -> 100k -> 270k -> -5V
-5V present: midpoint ~1.06V (below ref)
-5V absent: midpoint ~2.41V (above ref)
Reference: $3.3V * 82k / 182k = 1.49V$
OUT_A high (via R_PU) when -5V OK -> BIAS_OK HIGH -> Q1 ON
OUT_A low when -5V absent -> BIAS_OK LOW -> Q1 OFF (safe)



Thermal Monitor + Bias Sequencing Comparator
Private / Custom Drone Hardware

Sheet: /Control and Safety/
File: control.kicad_sch

Title: jamRF-v2 Control and Safety

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